to the plan output. Run the example above as `EXPLAIN (ANALYZE) SELECT 1;` to see Planning Time added to the output.

Reading query execution plans is a good skill to develop for multiple reasons. You'll also begin to understand how PostgreSQL parses a query and handles different pieces like selecting fields, filtering rows based on column criteria, or ordering results, and this can inform your query design and schema design.

You may have noticed the Planning Time is extremely short. To help make this possible, PostgreSQL relies on pre-calculated statistics about tables, indexes, and estimates. You may not have realized that PostgreSQL continually collects statistics about all the data in your database. You can even trigger manual statistics collection yourself.

Stats can be collected by running the `ANALYZE` command for a table. From psql, try running `ANALYZE trips;` to give this a spin.

It would be infeasible to run ANALYZE manually all the time. Fortunately PostgreSQL runs it automatically. Analyze runs automatically by being part of the Autovacuum process that is enabled by default in PostgreSQL.

When PostgreSQL is running, and Autovacuum is triggered for a table based on its trigger conditions, a `VACUUM (ANALYZE)` process runs for each table in the database. You haven't yet learned about the trigger thresholds and how to tune them, but you will later. In normal operations and for Rideshare, table statistics are updated automatically and the default values you have in your local development machine are fine for now.

Now you know how Stats are collected and you saw the `ANALYZE` argument with VACUUM.

`EXPLAIN` also supports an `ANALYZE` argument but it has a different purpose in that context.

EXPLAIN only shows a query execution plan but doesn't run it. To both show the execution plan and run the SQL query, use `EXPLAIN ANALYZE`.

With all of that in mind, it's time to start diving into query execution plans.

Query execution plans are packed with information. Besides the tables, indexes, and columns there are many more types of data. Some of the information is presented below. Some interpretation information and "next step" questions are listed above each type of item for you to begin thinking about. For now, focus more on seeing what kind of information is listed over the interpretation of the information.